

Requirement Analysis Report — SCRUM-101: E-commerce Checkout Process

Document Information

Field	Value
Requirement ID	SCRUM-101
Title	E-commerce Checkout Process
Document Type	User Story
Source File	C:\PlaywrightDemo2.0\user_stories\SCRUM-101-ecommerce-checkout.md
Application Under Test	https://www.saucedemo.com
Test Credentials	standard_user / secret_sauce
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Report Version	1.0

Business Objective

Enable a logged-in customer to complete a purchase on the SauceDemo store through an end-to-end checkout process: review the cart, enter checkout/shipping information, review the order (payment and shipping summary, subtotal/tax/total), and confirm the order. The story states the process must be intuitive, secure, and provide clear feedback at each step.

Requirement Summary

The story defines a 4-stage checkout flow (Cart Review -> Checkout Information Entry -> Order Overview -> Order Completion) plus a cross-cutting error-handling requirement, on the SauceDemo demo storefront. It is expressed as 5 acceptance criteria (AC1-AC5) and 5 business rules (BR1-BR5), with supporting technical notes on automation scope (Playwright, 3 browsers, mobile responsiveness) and a Definition of Done.

- Acceptance Criteria: 5
- Business Rules: 5
- Functional Requirements extracted: 11
- Non-Functional Requirements extracted: 4

Functional Requirements

ID	Description	Source
FR-01	The cart page must display all items added to the cart, including name, description, price, and quantity for each item.	AC1
FR-02	The cart page must display the total price calculation for the items in the cart.	AC1

ID	Description	Source
FR-03	The cart page must provide an option to continue shopping and an option to proceed to checkout.	AC1
FR-04	Clicking the "Checkout" button on the cart page must redirect the user to the checkout information page.	AC2
FR-05	The checkout information page must present form fields for First Name, Last Name, and Zip/Postal Code, all of which are mandatory.	AC2, BR1
FR-06	If any mandatory field on the checkout information page is left empty and the user clicks Continue, an error message indicating which field is required must be displayed.	AC2
FR-07	After entering valid checkout information and clicking "Continue", the user must be redirected to the checkout overview page.	AC3
FR-08	The checkout overview page must display a summary of all items in the order, payment and shipping information, and the subtotal, tax, and total amount.	AC3
FR-09	The checkout overview page must provide options to Cancel or Finish the order.	AC3, BR5 (partial)
FR-10	Clicking "Finish" on the checkout overview page must redirect the user to an order confirmation page showing a success message and a "Back Home" button that returns the user to the products page.	AC4
FR-11	Entering invalid data (e.g., special characters, incomplete information) on the checkout information page must produce appropriate validation error messages and prevent the user from proceeding until all fields are valid.	AC5

Non-Functional Requirements

ID	Category	Description	Source	Notes
NFR-01	Compatibility	The checkout flow must be tested across Chrome, Firefox, and Safari browsers.	Technical Notes	No specific browser versions specified.
NFR-02	Responsiveness	The checkout flow must be mobile responsive.	Technical Notes	No specific device list, breakpoints, or viewport sizes specified.
NFR-03	Usability	The checkout process should be intuitive, with clear feedback provided at each step.	Story Description	Not objectively measurable as stated; no concrete usability acceptance criteria or metric provided. Flagged as ambiguous.
NFR-04	Security	The checkout process should be secure.	Story Description	No concrete security requirement, data-handling rule, or acceptance criterion is defined anywhere in the document. Flagged as an ambiguous requirement gap.

Actors

Role	Identifier	Description
Customer / Logged-in User	standard_user	The only actor referenced in the document. Must be authenticated before checkout can be accessed (BR2). No other roles (e.g., guest, admin, locked-out user) are described in this story.

Acceptance Criteria

AC1: Cart Review

- GIVEN I am a logged-in user with items in my cart
- WHEN I navigate to the cart page
- THEN I should see all added items with their details (name, description, price, quantity)
- AND I should see the total price calculation
- AND I should have options to continue shopping or proceed to checkout

AC2: Checkout Information Entry

- GIVEN I am on the cart page with items
- WHEN I click the "Checkout" button
- THEN I should be redirected to the checkout information page
- AND I should see form fields for First Name, Last Name, and Zip/Postal Code
- AND all fields should be mandatory
- WHEN I leave any field empty and click Continue
- THEN I should see an error message indicating which field is required

AC3: Order Overview

- GIVEN I have entered valid checkout information
- WHEN I click the "Continue" button
- THEN I should be redirected to the checkout overview page
- AND I should see a summary of all items in my order
- AND I should see payment and shipping information
- AND I should see the subtotal, tax, and total amount
- AND I should have options to Cancel or Finish the order

AC4: Order Completion

- GIVEN I am on the checkout overview page
- WHEN I click the "Finish" button
- THEN I should be redirected to the order confirmation page
- AND I should see a success message confirming my order
- AND I should see a "Back Home" button to return to the products page

AC5: Error Handling

- GIVEN I am on the checkout information page
- WHEN I enter invalid data (e.g., special characters, incomplete information)
- THEN I should see appropriate validation error messages
- AND I should not be able to proceed until all fields are valid

Business Rules

ID	Rule	Linked Acceptance Criteria
BR1	All checkout form fields are mandatory.	AC2
BR2	Users must be logged in to access checkout.	AC1
BR3	Cart cannot be empty when proceeding to checkout.	(none)

ID	Rule	Linked Acceptance Criteria
BR4	Order confirmation should clear the cart.	(none)
BR5	Users can cancel checkout at any step and return to cart.	AC3

Dependencies

ID	Description	Type
DEP-01	A working login/authentication feature is required before checkout can be exercised (BR2), but login functionality itself is not specified in this document.	Feature dependency
DEP-02	The cart must already contain items before checkout begins (AC1, BR3); the add-to-cart feature that populates the cart is not specified in this document.	Feature dependency
DEP-03	Availability and stability of the external, third-party-hosted application at https://www.saucedemo.com is required for both manual and automated testing.	Environment dependency
DEP-04	Validity of the supplied test credentials (<code>standard_user</code> / <code>secret_sauce</code>) on the target environment.	Test data dependency
DEP-05	Cross-browser (Chrome, Firefox, Safari) and mobile-responsive test execution depends on availability of corresponding browser/device infrastructure.	Infrastructure dependency

Assumptions

ID	Description	Status
ASM-01	The document does not state any assumptions explicitly. The only inference drawn without altering scope is that "logged-in user" in AC1 refers to successful authentication using the supplied <code>standard_user</code> credentials, since no other login mechanism or role is described.	Requires confirmation

Risks

ID	Description	Impact	Related Item
RISK-01	Security is referenced in the story description ("secure") but no security acceptance criteria, business rule, or NFR detail is defined anywhere in the document, so security cannot be objectively verified.	High	NFR-04
RISK-02	BR4 ("Order confirmation should clear the cart") is not reflected in any acceptance criterion (AC4 only mentions a success message and Back Home button), so this business rule risks being untested if not explicitly added to the test plan.	Medium	BR4
RISK-03	BR5 ("Users can cancel checkout at any step and return to cart") is only partially reflected in the acceptance criteria (a Cancel option is described on the overview page in AC3, but not on the checkout information page), risking incomplete coverage of the business rule.	Medium	BR5
RISK-04	The story description states the customer will "select payment method," but AC3 only requires the overview page to display payment and shipping information, with no acceptance criterion describing a payment method selection step. SauceDemo's actual checkout has no payment method selection UI. This is a conflict between the narrative description and the acceptance criteria / actual application behavior.	Medium	AC3

ID	Description	Impact	Related Item
RISK-05	NFR-01 and NFR-02 require cross-browser and mobile-responsive testing but do not specify browser versions, device list, or viewport breakpoints, which can lead to inconsistent or incomplete coverage across test executions.	Low	NFR-01, NFR-02
RISK-06	The application under test is an external third-party demo site (saucedemo.com) not controlled by the project team; behavior changes or downtime are outside the team's control and could invalidate test results.	Low	Application URL

Edge Cases

ID	Description	Derived From
EDGE-01	Submitting the checkout information form with all fields empty (not just a single field) — AC2 only describes leaving "any field" empty; combined/multi-field empty submission behavior is not explicitly specified.	AC2
EDGE-02	Submitting special characters, leading/trailing whitespace, or excessively long values in First Name, Last Name, or Zip/Postal Code fields — AC5 references "special characters" and "incomplete information" generically without enumerating specific invalid input classes.	AC5
EDGE-03	Zip/Postal Code format validation (numeric-only vs. alphanumeric postal codes, e.g. international formats) is not defined.	AC2, AC5
EDGE-04	Behavior when attempting to reach the checkout information or overview page directly via URL without following the prescribed flow (e.g., with an empty cart, or without being logged in) is not defined.	BR2, BR3
EDGE-05	Behavior of the browser Back button during the checkout flow is referenced only in Technical Notes ("Test navigation flow and back button behavior") without a corresponding acceptance criterion describing expected behavior.	Technical Notes
EDGE-06	Single-item vs. multi-item cart checkout, and a cart with a large/maximum quantity of items, are not distinguished in any acceptance criterion.	AC1

Open Questions

ID	Question	Related Item
OQ-01	What is the exact expected error message text/format for each mandatory field left empty in AC2 (e.g., does SauceDemo's actual "Error: First Name is required" style message match the expected wording)?	AC2, FR-06
OQ-02	What specific characters or input patterns constitute "invalid data" in AC5? The document gives examples (special characters, incomplete information) but no exhaustive or authoritative validation rule set.	AC5, FR-11
OQ-03	Is BR4 (cart clearing on order confirmation) an in-scope, testable requirement even though no acceptance criterion covers it? Should an acceptance criterion be added, or is it intentionally out of scope for this story?	BR4
OQ-04	Does BR5 ("cancel checkout at any step") apply to the checkout information page (AC2) as well as the overview page (AC3), given AC2 does not mention a Cancel option?	BR5, AC2

ID	Question	Related Item
OQ-05	The story description mentions the customer will "select payment method," but no acceptance criterion describes a payment method selection step. Is this narrative aspirational/out of scope for SauceDemo, or is an acceptance criterion missing?	Story Description, AC3
OQ-06	What concrete, measurable security requirements apply to the checkout process (data handling, HTTPS enforcement, session handling, etc.)? None are specified.	NFR-04
OQ-07	Which specific browser versions and mobile devices/viewport sizes should be covered for NFR-01 and NFR-02?	NFR-01, NFR-02
OQ-08	Are there any other actors/roles besides <code>standard_user</code> (e.g., <code>locked_out_user</code> , <code>problem_user</code> , <code>performance_glitch_user</code> , which exist on SauceDemo) that should be in scope for this story?	Actors

Requirement Traceability

Requirement ID	Acceptance Criteria	Business Rules
FR-01	AC1	—
FR-02	AC1	—
FR-03	AC1	—
FR-04	AC2	—
FR-05	AC2	BR1
FR-06	AC2	BR1
FR-07	AC3	—
FR-08	AC3	—
FR-09	AC3	BR5
FR-10	AC4	BR4
FR-11	AC5	—
NFR-01	—	—
NFR-02	—	—
NFR-03	—	—
NFR-04	—	—
BR2 (login required)	AC1	BR2
BR3 (cart not empty)	—	BR3

QA Coverage Recommendations

1. Cover each acceptance criterion (AC1-AC5) with at least one positive/happy-path test case and, where applicable, negative test cases.
2. Add explicit test coverage for BR3 (attempting checkout with an empty cart) and BR4 (verifying the cart is cleared after order confirmation), since neither is covered by an existing acceptance criterion — flag to product owner as a gap to confirm

before automating.

3. Add explicit test coverage for cancel behavior on the checkout information page (AC2) in addition to the overview page (AC3), to fully validate BR5.
4. Design boundary and negative test cases for the mandatory fields in AC2/AC5: empty field, whitespace-only, special characters, very long input, and (for Zip/Postal Code) non-numeric/alphanumeric formats.
5. Verify the exact error message text for each individual required field, and for multi-field-empty submission.
6. Include cross-browser regression coverage (Chrome, Firefox, Safari) per NFR-01 for the full checkout flow, at minimum on the latest stable version of each browser pending clarification (OQ-07).
7. Include mobile-responsive checkout coverage per NFR-02 at common breakpoints (e.g., mobile portrait/landscape, tablet) pending clarification of the target device list (OQ-07).
8. Verify browser Back-button behavior during each step of the checkout flow, per the Technical Notes, even though no acceptance criterion formally defines expected behavior (EDGE-05) — flag to product owner.
9. Do not author security-specific automated assertions until concrete security requirements are defined (OQ-06); flag this as an untestable NFR in its current form.
10. Confirm with the product owner whether payment-method selection (mentioned in the story description) is in scope; if not, no test case should assert its presence, since AC3 does not require it.
11. Include a full end-to-end happy-path scenario chaining AC1 through AC4 (cart -> info -> overview -> confirmation) as the primary regression test.
12. Include multi-item cart scenarios (not just single-item) to validate totals, tax, and subtotal calculations in AC1 and AC3.